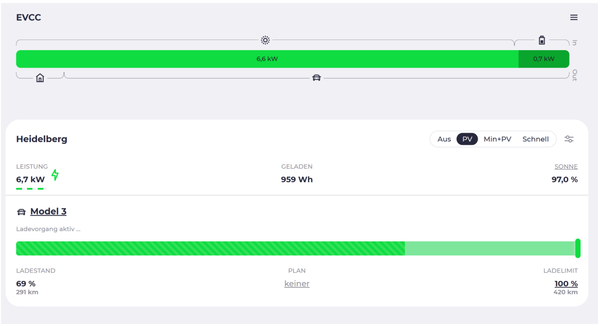
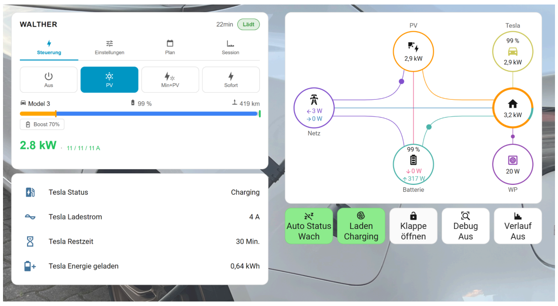
Charge-on-solar for Tesla:

EVCC with Home Assistant and Bluetooth via ESP32 (without cloud)





1. Installing the ESP32:

There are separate instructions for this:https://docs.google.com/document/d/1W33jPQH4uAfSzb8rHvg7sBFm9zY_D-149qrtlnRN2xQ/edit?usp=sharing

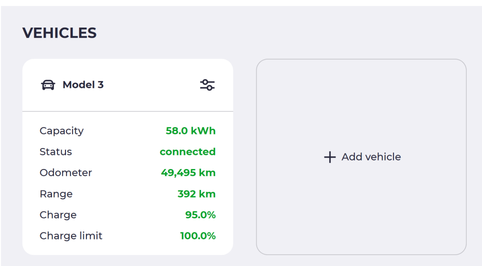
Attention: In Chapter 4: In the configuration, the # characters must be removed from the package evcc.yml:

```
packages:
  remote_package_files:
    url: https://github.com/PedroKTFC/esphome-tesla-ble/
    files:
      - packages/base.yml
      - packages/common.yml
      - packages/project.yml
      - packages/client.yml
  #    - packages/listener.yml  # Uncomment this to scan find your VIN BLE MAC address
      - packages/evcc.yml  # Uncomment this if you use the ESP for EVCC charge control
```

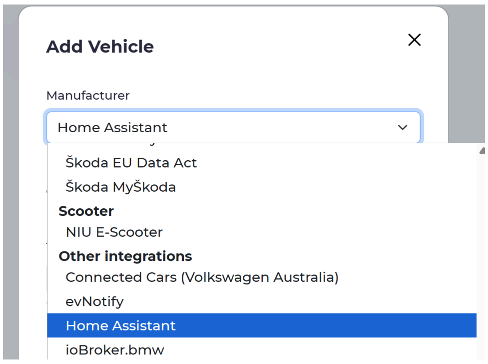
2. Connect EVCC

2.1. Setting up the car

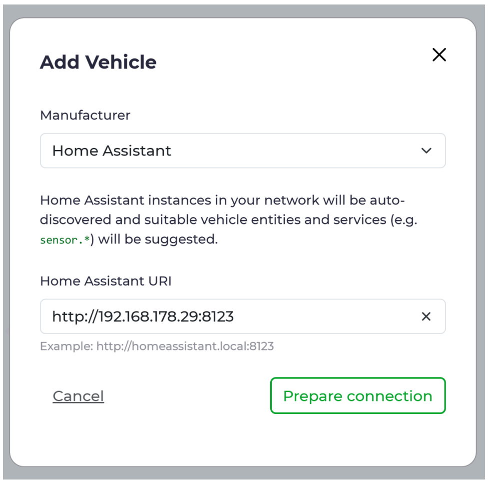
In EVCC, click on More/Configuration, then on “Add vehicle”:



We then select “Home Assistant” as the manufacturer (far down):



Next, the Home Assistant server is specified. It is usually found automatically; otherwise, the server's IP address is entered here.



Now click “Prepare connection”. The first time you do this, a Home Assistant login window will appear, where you need to enter a valid HA username and password. Don't worry, the username is not stored in EVCC, only a “persistent login token”.

From here on, sensors need to be entered. Vehicle name and battery capacity are entered manually. I've allocated 2 kWh of degradation from the nominal capacity. The nominal value is also quite accurate.

Edit Vehicle

Manufacturer

Home Assistant

Home Assistant instances in your network will be auto-discovered and suitable vehicle entities and services (e.g. `sensor.*`) will be suggested.

Title

Model 3

Will be displayed in the user interface

Icon

change

Will be displayed in the user interface

Home Assistant URI

http://192.168.178.29:8123

Example: http://homeassistant.local:8123

State of charge [%]

sensor.tesla_ble_f549c4_charge_level

Example: sensor.vehicle_soc
Entity ID for the vehicle's battery state of charge in percent

Battery capacity optional

58 kWh

Example: 50

Remaining range [km] optional

sensor.tesla_ble_f549c4_range

Example: sensor.vehicle_range
Entity ID for the vehicle's remaining range in kilometers

Charging status optional

sensor.tesla_ble_f549c4_charging_state

Example: sensor.vehicle_charging
Entity ID for charging status (A=disconnected, B=connected, C=charging)

Target state of charge [%] optional

number.tesla_ble_f549c4_charging_limit

Example: number.vehicle_target_state_of_charge
Entity ID for the vehicle's target state of charge in percent (`number` or `input_number` entity)

Odometer [km] optional

sensor.tesla_ble_f549c4_odometer

Example: sensor.vehicle_odometer
Entity ID for the vehicle's odometer reading in kilometers

Climatisation active optional

binary_sensor.tesla_ble_f549c4_climate

Example: binary_sensor.vehicle_climater
Entity ID for the vehicle's climatisation state (`binary_sensor` with `on/off` state)

Finish time optional

sensor.tesla_ble_f549c4_minutes_to_limit

Example: sensor.vehicle_finish_time
Entity ID for the estimated charging finish time (ISO8601 or Unix timestamp)

Service to start charging optional

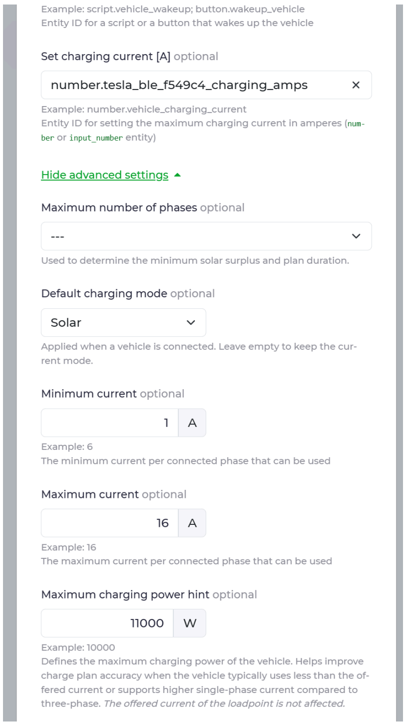
switch.tesla_ble_f549c4_charger

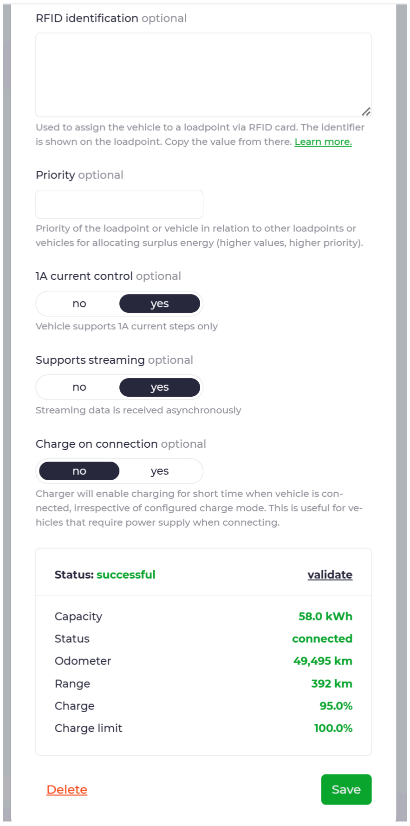
Example: script.vehicle_start_charge; switch.vehicle_charger
Entity ID for a script or a switch that starts charging the vehicle. Only useful with the docs.evcc.io. If a switch is provided, Service to stop charging must be left empty.

Service to stop charging optional

Example: script.vehicle_stop_charge
Entity ID for a script that stops charging the vehicle. Only useful with the docs.evcc.io.

Service to wake up vehicle optional



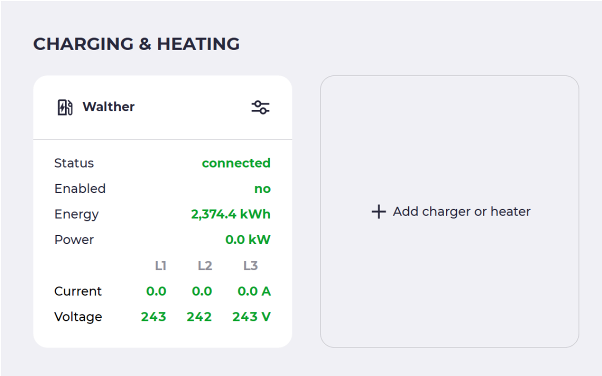


The number of phases and the current must be specified manually. The absolute minimum for Tesla is 1A, but for efficiency reasons I use 2A.

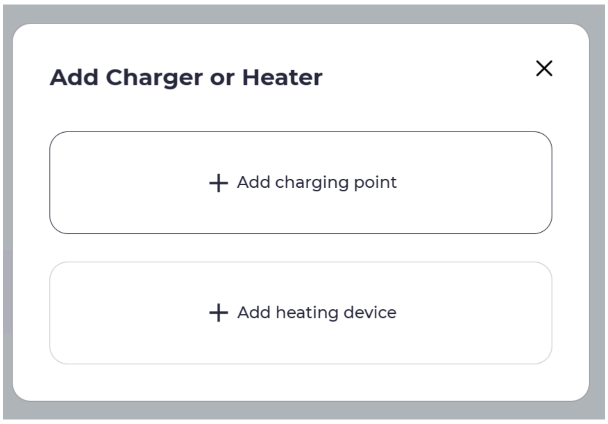
Finally, click “validate”. Plausible values must be displayed when the car is within range. Then, click “Save”.

2.2. Setting up the wallbox:

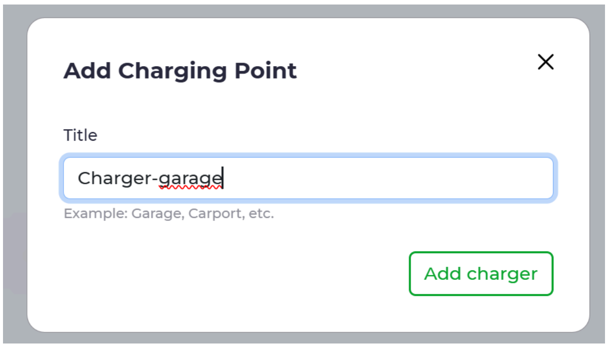
Click “Add charger or heater”:



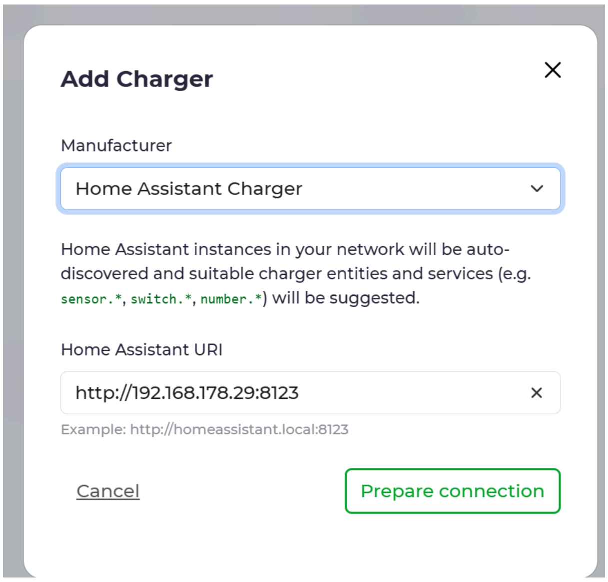
Then click “Add charging point”:



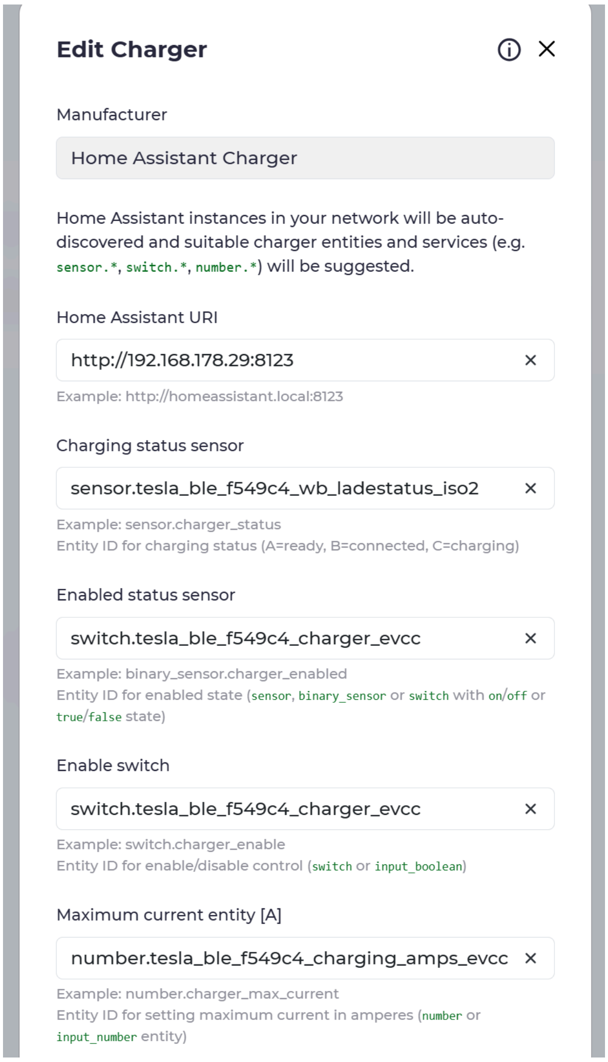
Assign a name (e.g., manufacturer name, garage left, carport,...) and select “Add charger”



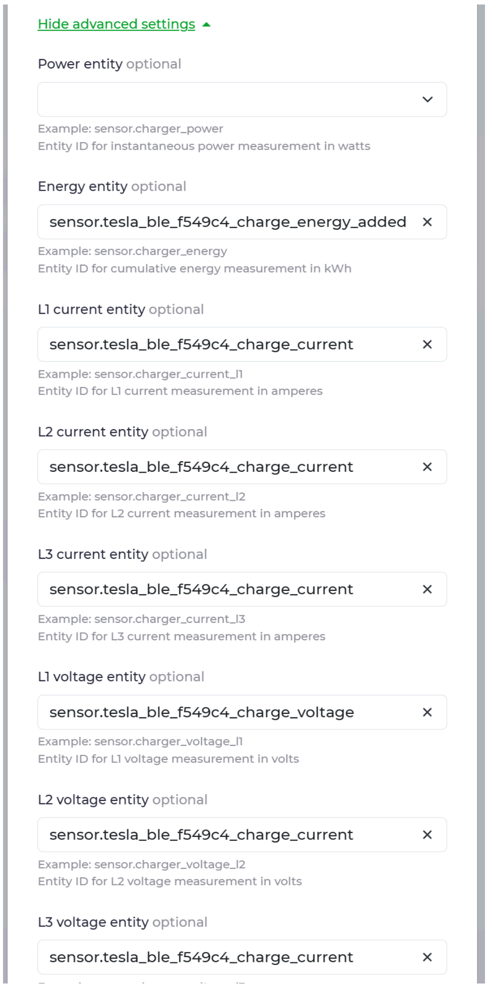
Select “Home Assistant Charger” again (this time near the top) and click “Prepare connection”:



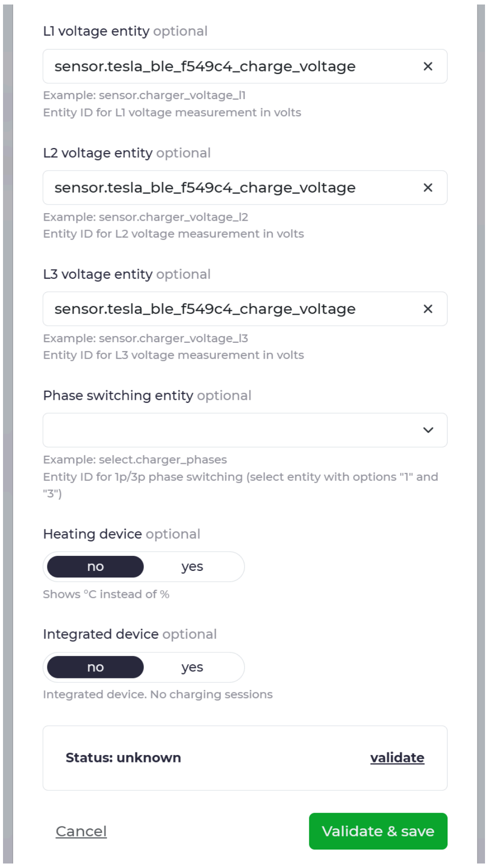
The entities are again selected via autocomplete:

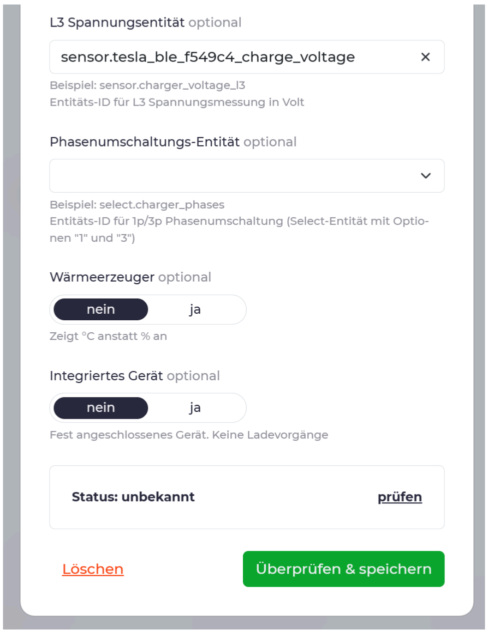


Moving on – the power output is intentionally left blank here! The car provides a value, but it's rounded to the nearest kW, which is very inaccurate. EVCC can calculate the power itself from voltage and current. Notice: The current and voltage sensors are entered thrice on 3-phase-systems. On single- or dual-phase systems, leave the unused sensors empty.

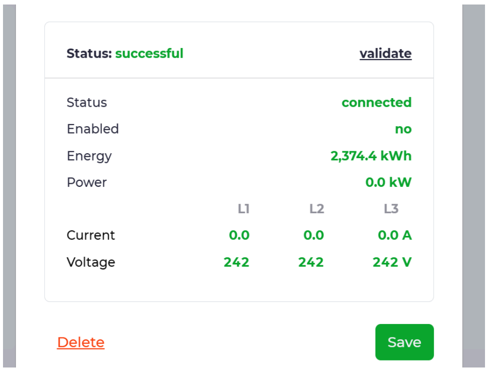


The last entities:





After clicking "Check", values should reappear. It's normal for very low voltage values to appear; expect 2-6V when you are not charging, and normal 115 or 230V when charging.

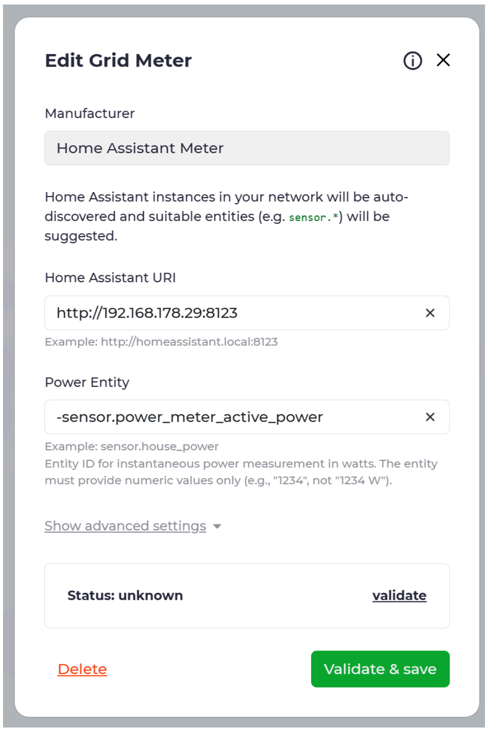


Click “Save”.

3. PV system configuration

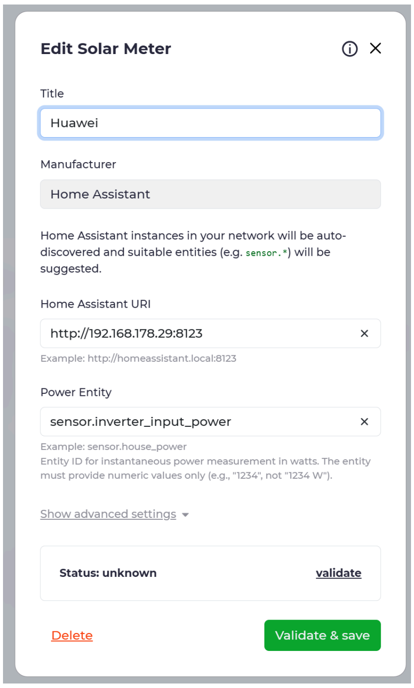
EVCC can query most PV systems directly; alternatively, the data can also be retrieved via Home Assistant. For my Huawei system, there's a very comprehensive integration that queries the system via Modbus/TCP every 15 seconds. Some might use a Modbus proxy, which involves a lot of effort and querying the data a second time. It's simpler to just let Home Assistant do the work and retrieve the data from there. Regarding energy data, it's important to note that EVCC always expects positive values when power or energy flows towards the vehicle, e.g., from the roof to the home network, from the home battery to the home network, or from the grid to the home network. The opposite direction (feed-in or charging the home battery) must be negative. If the integration provides the values in reverse, add a hyphen ("-") before the sensor name; this reverses the measurement direction for EVCC.

3.1 Network meter

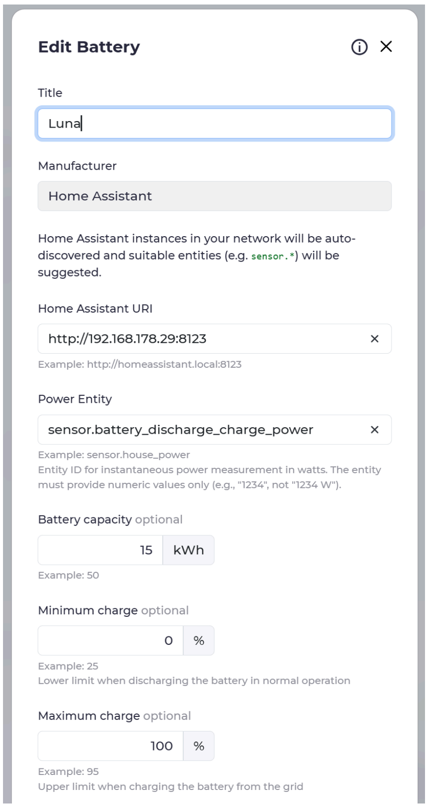


The grid meter for my Huawei system is a case in point: grid feed-in is positive, grid consumption is negative. That's why there's a "-" before the sensor name. Note: I've seen the opposite meter direction with Huawei devices using Emma. In that case, the "-" is omitted.

3.2 Inverter power meter:



3.3 Home battery:



4. Finish!

Once the PV system is registered, EVCC is functional; nothing more is needed. Optionally, you can also set up the EVCC integration from Marq24, which allows you to display EVCC data, for example, in dashboards.

https://github.com/marq24/ha-evcc

And there are some nice maps for Home Assistant from mkshb:

https://github.com/mkshb/hass-evcc-card